

CATANIA SIGONELLA, Italy

WMO: 164590

Lat: 37.406N

Lon: 14.919E

Elev: 72

StdP: 14.66

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	35.3	37.5	24.8	18.9	54.8	28.7	22.7	52.4	31.1	55.8	28.0	55.0	6.8	270	0.490

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	23.5	98.5	71.9	95.1	71.9	92.8	71.8	79.7	84.9	78.1	83.8	76.5	82.8	13.4	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
78.7	149.5	81.7	76.8	140.2	80.1	75.0	131.8	79.1	43.4	85.8	41.5	84.7	40.0	84.3	90.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
26.9	23.3	20.8	30.2	106.5	2.4	4.0	28.5	109.4	27.1	111.7	25.7	113.9	24.0	116.8	
			DB												
			WB	29.8	82.9	2.2	2.1	28.2	84.4	27.0	85.6	25.7	86.8	24.1	88.3

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	64.5	50.5	50.9	54.4	59.1	66.5	74.8	80.1	80.9	75.3	68.0	59.5	52.9
	DBStd	11.72	4.07	4.17	4.32	3.98	4.41	4.90	3.85	3.48	4.51	4.66	4.84	4.32
	HDD50	111	42	36	10	0	0	0	0	0	0	0	2	21
	HDD65	1964	450	396	330	181	33	1	0	0	1	23	173	376
	CDD50	5398	57	60	146	273	512	744	934	959	758	559	286	110
	CDD65	1775	0	0	1	4	80	295	469	494	308	117	7	0
	CDH74	18324	0	0	21	66	776	3131	5588	5596	2474	637	34	1
	CDH80	8393	0	0	6	6	191	1367	2921	2862	916	123	1	0
Wind	WSAvg	9.3	9.1	9.8	9.9	10.2	10.1	10.0	9.6	8.9	8.8	8.2	8.5	8.7
Precipitation	PrecAvg	20.4	2.7	1.9	1.6	1.6	0.9	0.4	0.2	0.6	1.1	3.7	2.4	3.1
	PrecMax	36.0	8.9	6.6	5.3	6.8	5.3	1.5	1.3	3.9	3.3	16.9	9.1	13.1
	PrecMin	6.6	0.2	0.2	0.2	0.2	0.0	0.0	0.0	0.0	0.1	0.0	0.0	0.1
	PrecStd	7.1	2.2	1.7	1.0	1.5	1.0	0.4	0.4	0.9	0.8	4.1	2.2	2.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	67.3	69.3	76.6	79.8	90.5	100.8	104.0	102.6	96.9	87.8	77.3	69.9
		MCWB	56.2	56.3	58.6	61.0	65.9	70.8	71.5	72.6	71.0	68.1	64.6	59.2
	2%	DB	64.3	64.8	71.2	75.4	85.5	94.6	98.5	98.3	91.6	82.8	74.6	66.3
		MCWB	54.0	53.6	56.6	59.2	64.6	69.7	71.7	72.8	73.1	69.2	63.5	57.2
	5%	DB	61.5	62.6	67.7	73.0	81.0	91.1	95.0	95.0	88.2	80.4	71.6	64.1
		MCWB	53.1	53.0	55.1	59.0	63.1	69.5	72.3	72.6	72.2	68.2	62.7	55.6
	10%	DB	59.2	60.4	64.4	69.6	78.4	87.7	92.4	92.2	85.6	77.2	68.4	61.2
		MCWB	52.3	52.4	54.1	57.9	62.3	68.1	71.9	72.6	70.9	67.8	61.4	55.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	60.1	60.5	61.9	66.3	71.5	78.2	81.0	83.3	81.3	78.0	70.7	63.1
		MCDB	62.8	62.5	68.8	72.5	79.6	84.5	88.2	87.9	85.4	80.1	73.3	65.5
	2%	WB	57.4	57.7	59.7	63.7	69.0	75.2	78.7	80.1	78.5	74.8	67.4	60.4
		MCDB	61.0	61.2	65.4	69.4	76.4	82.7	85.1	84.8	82.8	77.5	70.4	63.5
	5%	WB	55.6	55.8	58.0	61.8	67.2	73.3	76.8	78.4	76.4	72.2	65.1	58.4
		MCDB	59.0	59.6	63.0	67.3	74.5	82.4	84.1	84.3	81.2	75.7	68.7	61.4
	10%	WB	53.9	54.0	56.5	60.1	65.5	71.6	75.2	76.9	74.6	70.1	63.1	56.6
		MCDB	57.8	58.3	61.2	65.6	72.8	81.2	83.4	83.3	79.8	74.0	66.7	59.8
Mean Daily Temperature Range	5% DB	MDBR	17.9	18.6	20.0	20.6	22.5	23.8	24.5	23.5	20.1	18.7	18.1	17.3
		MCDBR	21.3	21.9	25.5	26.2	27.2	28.4	28.9	27.4	24.0	22.0	21.2	21.0
		MCWBR	13.3	12.8	13.3	13.0	11.0	11.5	11.7	11.6	10.8	11.4	12.5	12.8
	5% WB	MCDBR	18.1	18.4	20.8	20.8	22.2	24.5	24.1	22.5	21.0	18.2	17.8	16.9
		MCWBR	13.1	12.4	13.2	12.8	11.6	12.1	12.4	12.1	11.4	12.1	12.8	12.5
Clear-Sky Solar Irradiance	taub	0.356	0.382	0.433	0.474	0.476	0.459	0.458	0.467	0.464	0.435	0.394	0.362	
	taud	2.383	2.287	2.134	2.045	2.086	2.176	2.194	2.168	2.179	2.227	2.304	2.376	
	Ebn at Noon	259	266	261	257	259	263	262	257	249	244	241	247	
	Edh at Noon	29	36	45	53	51	47	46	46	43	37	31	27	
All-Sky Solar Radiation	RadAvg	726	974	1372	1746	2088	2339	2410	2116	1578	1144	801	647	
	RadStd	60	91	105	122	115	110	76	118	83	84	61	56	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (8 neighbors)	N/A	N/A	N/A	N/A	-0.68	-0.88	N/A	N/A	N/A	N/A

Nomenclature: See separate page